

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks
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COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

Exp no : 01

Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

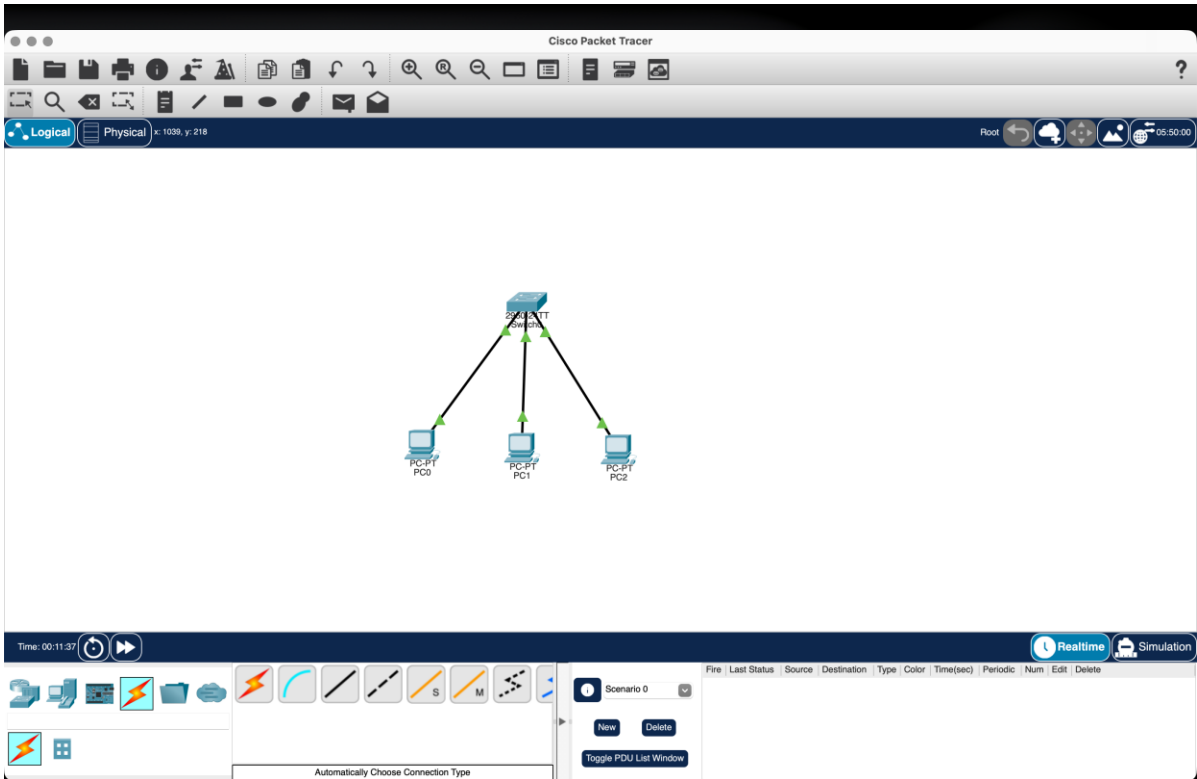
Aim:

To construct a Local Area Network (LAN) with multiple hosts connected through a switch using Cisco Packet Tracer, clear the ARP cache of a host, and use the ping command to initiate communication. The ARP Request and ARP Reply packets are captured and analyzed using Wireshark to understand how ARP resolves an IP address into the corresponding MAC address before data transmission.

Procedure:

1. Open Cisco Packet Tracer, add one switch and three PCs, and connect all PCs to the switch using Copper Straight-Through cables.
2. Configure the IP addresses of the PCs in the same network, for example 192.168.1.10, 192.168.1.20, and 192.168.1.30 with subnet mask 255.255.255.0.
3. Open the Command Prompt of one PC and check its ARP table using the arp -a command. Clear the ARP cache using the appropriate ARP command.
4. Start Wireshark packet capture and initiate communication from the selected PC to another PC using the ping command.
5. Apply the Wireshark filter arp and identify the ARP Request packet, where the sender broadcasts a request to find the MAC address associated with the target IP address.
6. Identify the ARP Reply packet, where the destination PC responds with its MAC address. Note the sender and target MAC/IP addresses in both packets.

- 7. Observe the subsequent ICMP Echo Request and Echo Reply packets to verify that communication between the two hosts is successful.
- 8. Analyze the captured packets and explain how ARP maps the logical IP address to the physical MAC address before the actual data transmission takes place.

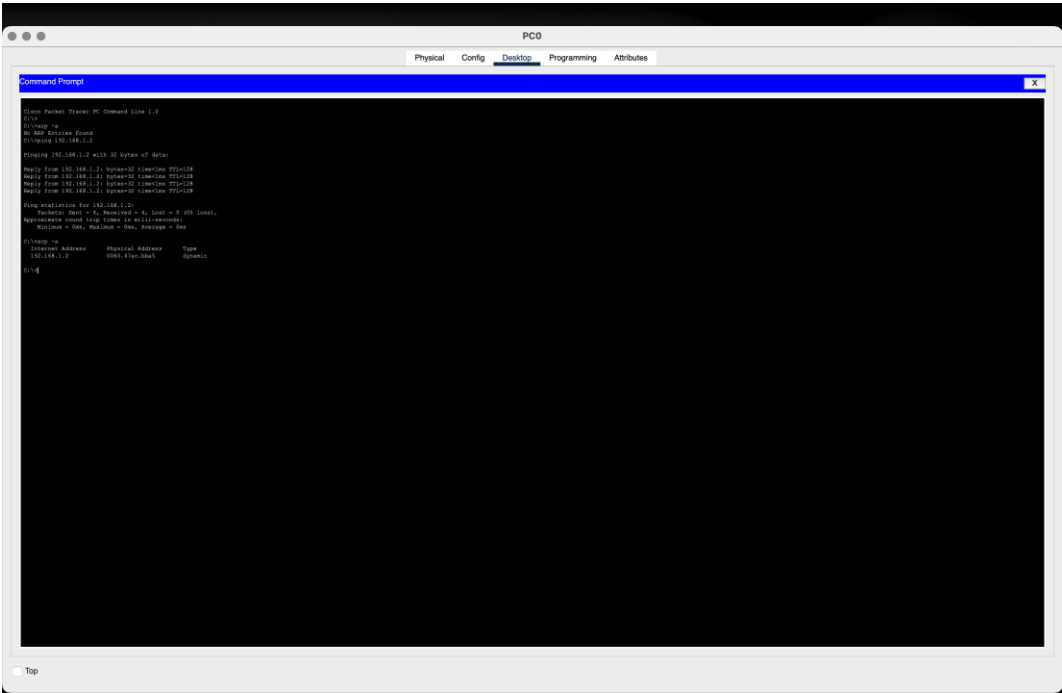


Implementation-1:
Fig1:LAN Topology with Multiple Hosts Connected Through a Switch

Explanation:

In Cisco Packet Tracer, **three PCs are connected to a single switch using Ethernet cables** to form a Local Area Network (LAN). Each PC is assigned a unique IP address within the same network, such as 192.168.1.1, 192.168.1.2, and 192.168.1.3, with the subnet mask 255.255.255.0. The switch provides communication between the connected hosts, allowing them to exchange data within the LAN. Connectivity is verified by using the ping command between the PCs.

S. No.	Device	IP Address	Subnet Mask	Purpose
1	PC0	192.168.1.1	255.255.255.0	Source host
2	PC1	192.168.1.2	255.255.255.0	Destination host
3	PC2	192.168.1.3	255.255.255.0	Additional LAN host
4	Switch	—	—	Connects all hosts



Implementation-2:

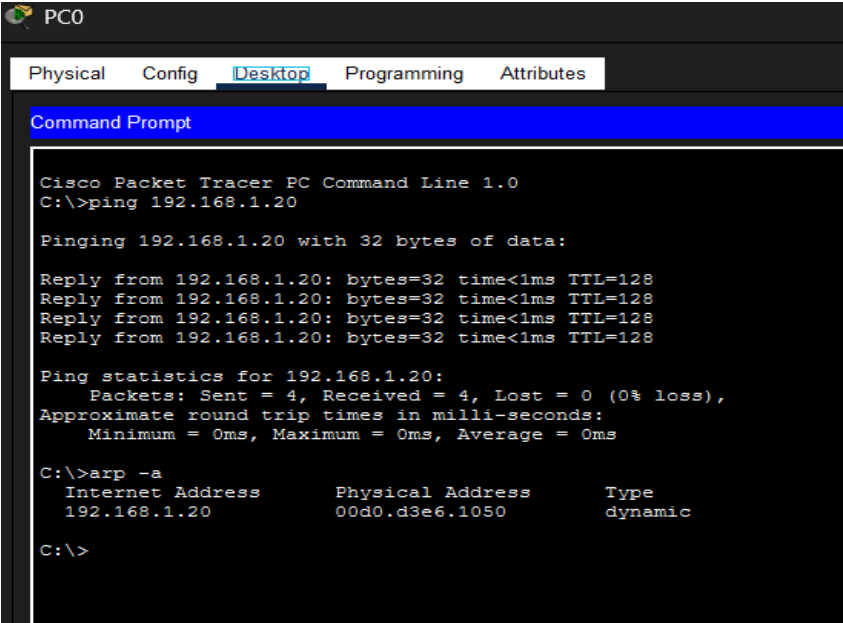


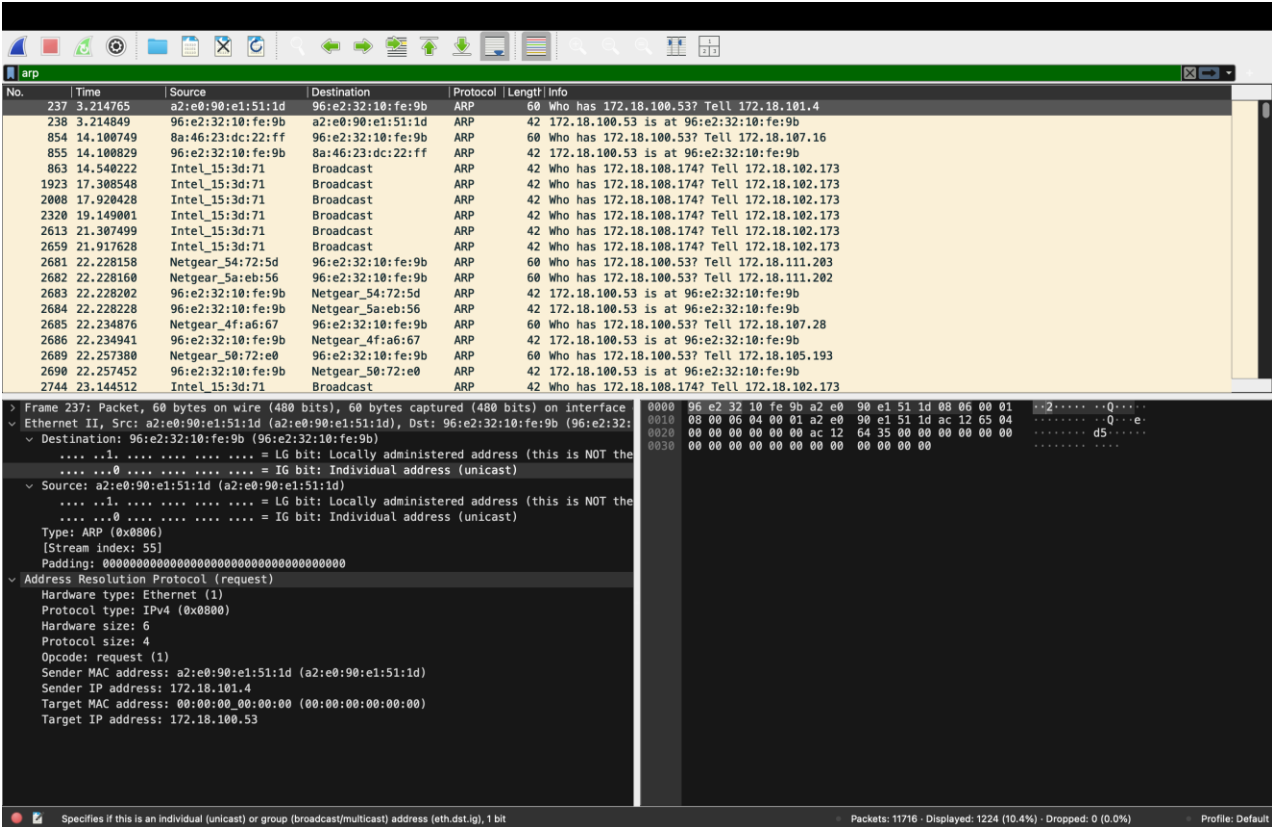
Fig2:ARPCache Verification and Ping Communication

Explanation:

The figure shows the **PC0 Command Prompt** in Cisco Packet Tracer. Initially, the ARP table is empty, as indicated by “**No ARP Entries Found.**” PC0 then sends a ping to PC1 at IP address **192.168.1.2**. The four successful replies confirm that communication between the two hosts is working correctly. After the ping, the `arp -a` command displays the dynamically learned MAC address corresponding to 192.168.1.2. This demonstrates that ARP successfully resolved the destination IP address into its MAC address for communication.

Parameter	Result
Source IP	192.168.1.1
Destination IP	192.168.1.2
Ping Packets Sent	4
Ping Packets Received	4
Packet Loss	0%

ARP Status Before Ping	No ARP Entries Found
ARP Status After Ping	MAC address learned dynamically
Communication Status	Successful



Implementation-3:
Fig3: ARP Request Packet Captured in Wireshark
Explanation:

The ARP Request is sent by the device with IP address **172.18.101.4**. Since it does not know the MAC address corresponding to **172.18.100.53**, it sends an ARP Request asking which device owns that IP address. The target MAC is shown as 00:00:00:00:00:00 because the MAC address is not yet known.

Field	Value
Packet Type	ARP Request
Sender MAC	a2:e0:90:e1:51:1d
Sender IP	172.18.101.4
Target MAC	00:00:00:00:00:00
Target IP	172.18.100.53

Implementation 4

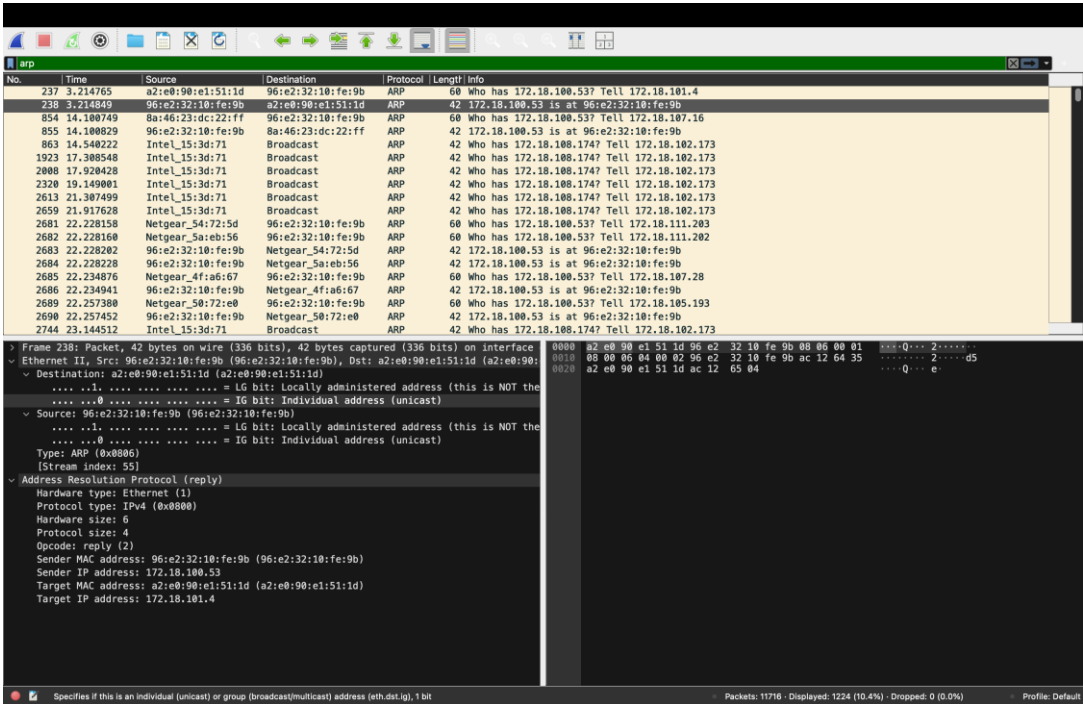


Figure 4: ARP Reply Packet Captured in Wireshark

Explanation

The ARP Reply is sent by the device with IP address **172.18.100.53** in response to the ARP Request. It provides its MAC address **96:e2:32:10:fe:9b** to the device with IP address **172.18.101.4**. This allows the requesting device to associate the destination IP address with its physical MAC address and proceed with data transmission.

Field	ARP Reply
Packet No.	238
Sender MAC	96:e2:32:10:fe:9b
Sender IP	172.18.100.53
Target MAC	a2:e0:90:e1:51:1d
Target IP	172.18.101.4

Analysis and Explanation of ARP:

From the captured Wireshark packets, the **ARP Request** shows that the sender IP address is **172.18.101.4** with MAC address **a2:e0:90:e1:51:1d**. The target IP address is **172.18.100.53**, while the target MAC address is initially unknown (00:00:00:00:00:00).

The **ARP Reply** is sent by **172.18.100.53**, whose MAC address is **96:e2:32:10:fe:9b**. It provides this MAC address to the sender **172.18.101.4**.

Result :

The LAN was successfully constructed using multiple hosts connected through a switch in Cisco Packet Tracer. Connectivity between the hosts was verified using the ping command. The ARP Request and ARP Reply packets were captured and analyzed using Wireshark, demonstrating how a host resolves an IP address to a MAC address before communication.

Experiment 02

2. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Aim:

To develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer, configure the required IP addresses, and verify connectivity using the ping command. The ICMP Echo Request and Echo Reply packets are captured and analyzed in Wireshark by examining fields such as Type, Code, Identifier, Sequence Number, and Round Trip Time (RTT).

Procedure:

1. Create a topology with two PCs and one router in Cisco Packet Tracer and connect them using Ethernet cables.
2. Configure IP addresses, subnet masks, and default gateways for both PCs and the router interfaces.
3. Use the ping command from one PC to the other to verify network connectivity.
4. Start Wireshark and capture the ICMP traffic generated by the ping operation.
5. Apply the icmp filter in Wireshark and identify the Echo Request (Type 8) and Echo Reply (Type 0) packets.
6. Examine the Type, Code, Identifier, Sequence Number, and packet timestamps to determine the RTT.
7. Analyze the results and explain how ICMP helps in connectivity verification and network troubleshooting.

Implementation 1

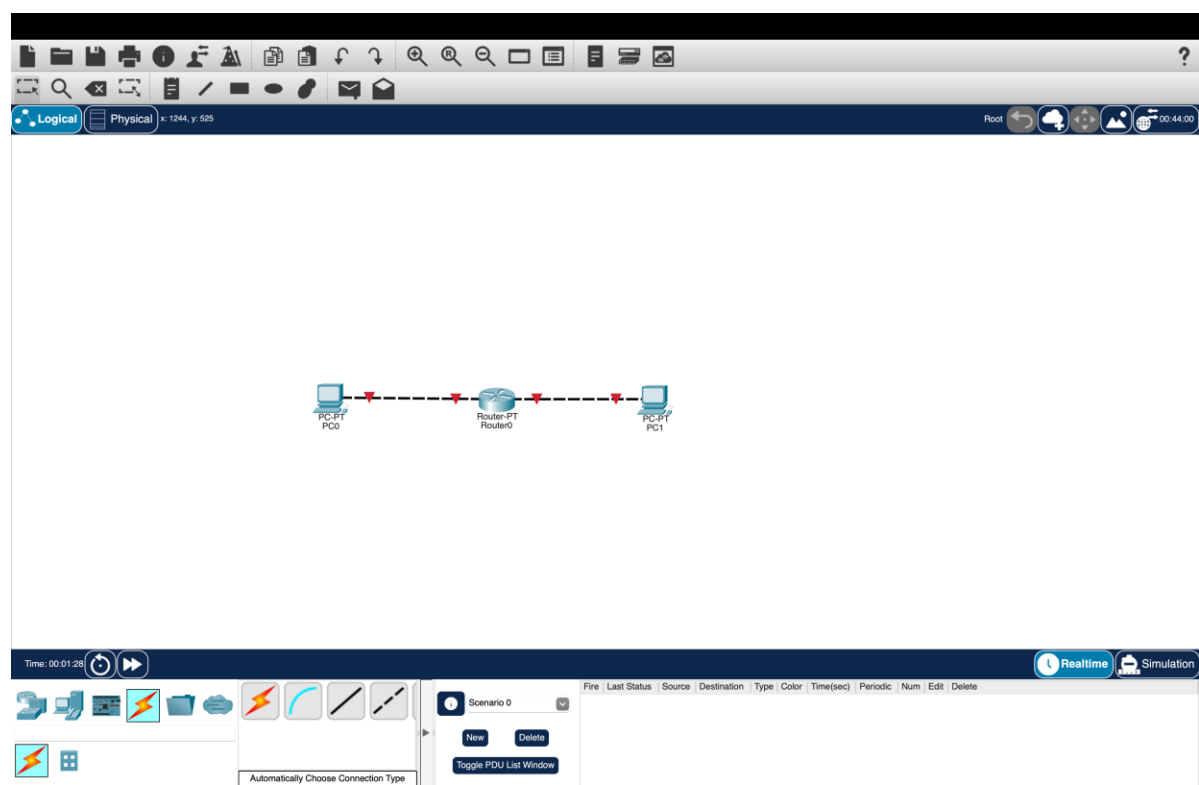


Fig1:Network Topology with Two PCs Connected Through a Router

Explanation:

The figure shows a simple network topology created in Cisco Packet Tracer, consisting of two PCs (PC0 and PC1) connected through a router (Router0). PC0 is connected to one interface of the router, while PC1 is connected to another interface. The router acts as an intermediary device that forwards data packets between the two PCs and their respective networks. The red link indicators show that the interfaces are currently down, so the router interfaces need to be configured and enabled before communication can take place.

Implementation-2:

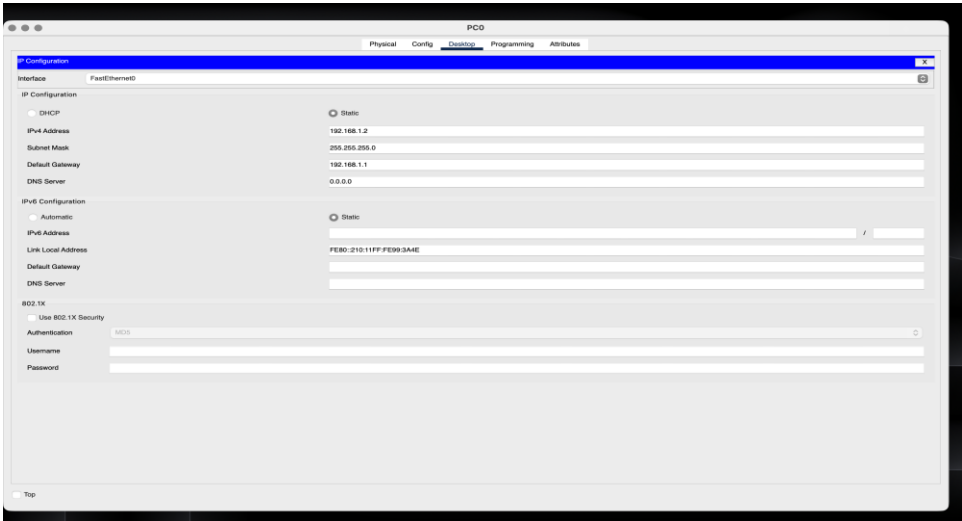


Fig2:IPAddress Configuration of PC0

Explanation:

The figure shows the IPv4 configuration of PC0 in Cisco Packet Tracer. A static IP address of 192.168.1.2 is assigned with the subnet mask 255.255.255.0. The default gateway is configured as 192.168.1.1, which is the router interface connected to PC0. This configuration allows PC0 to communicate with the router and access devices on other networks through the router.

Implementation-3:

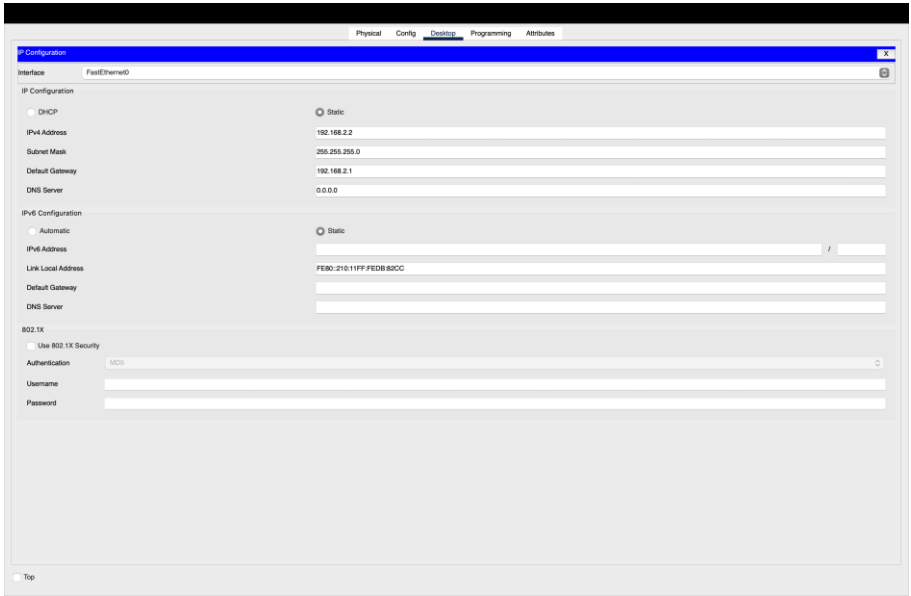
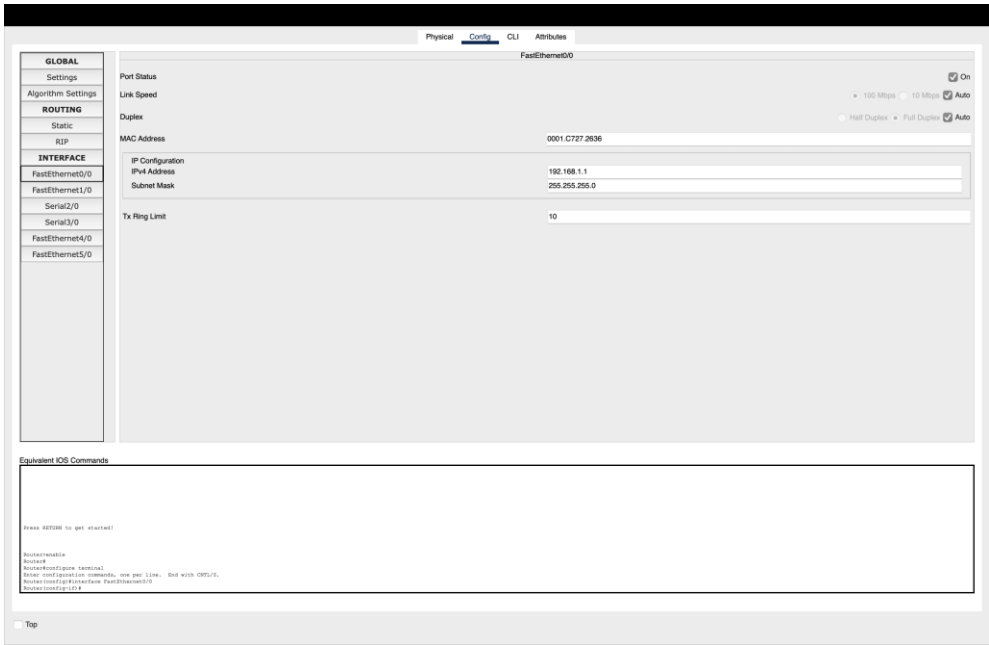


Figure 3: IP Address Configuration of PC1

Explanation:

The figure shows the IPv4 configuration of PC1 in Cisco Packet Tracer. PC1 is assigned the static IP address 192.168.2.2 with the subnet mask 255.255.255.0. The default gateway is configured as 192.168.2.1, which is the router interface connected to PC1. This configuration allows PC1 to communicate with the router and with devices on other networks.

Implementation-4:



Configuration for PC0 Network
Explanation:

The figure shows the configuration of **FastEthernet0/0** on Router0. The interface is enabled and assigned the IP address **192.168.1.1** with subnet mask **255.255.255.0**. This interface acts as the **default gateway for PC0 (192.168.1.2)** and connects PC0's network to the router.

Implementation-5:

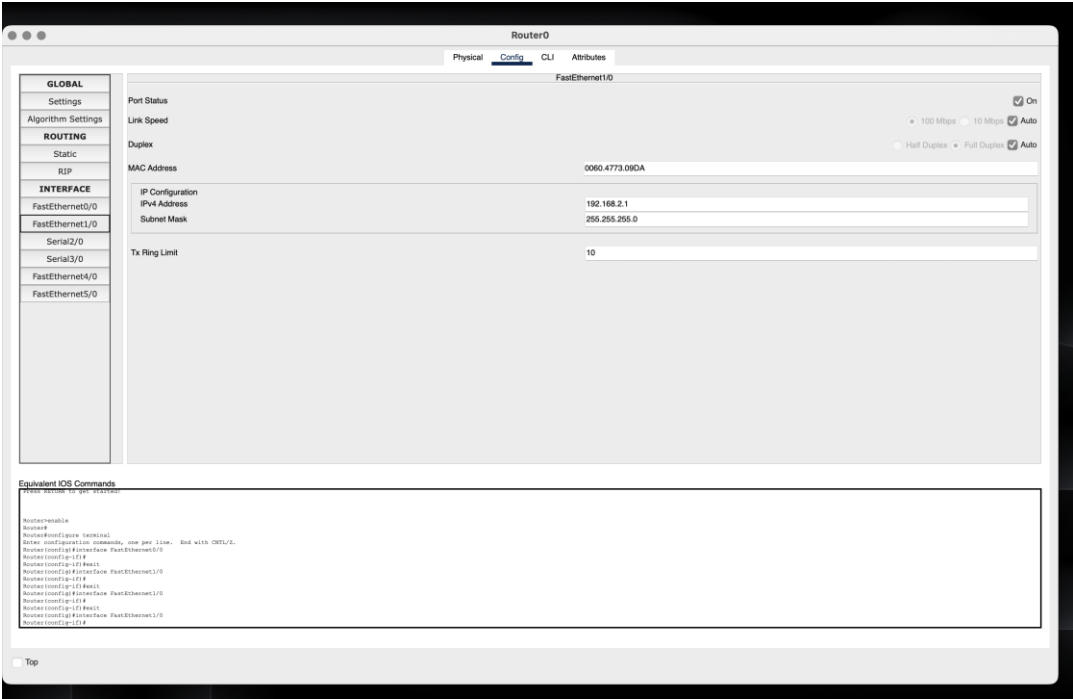
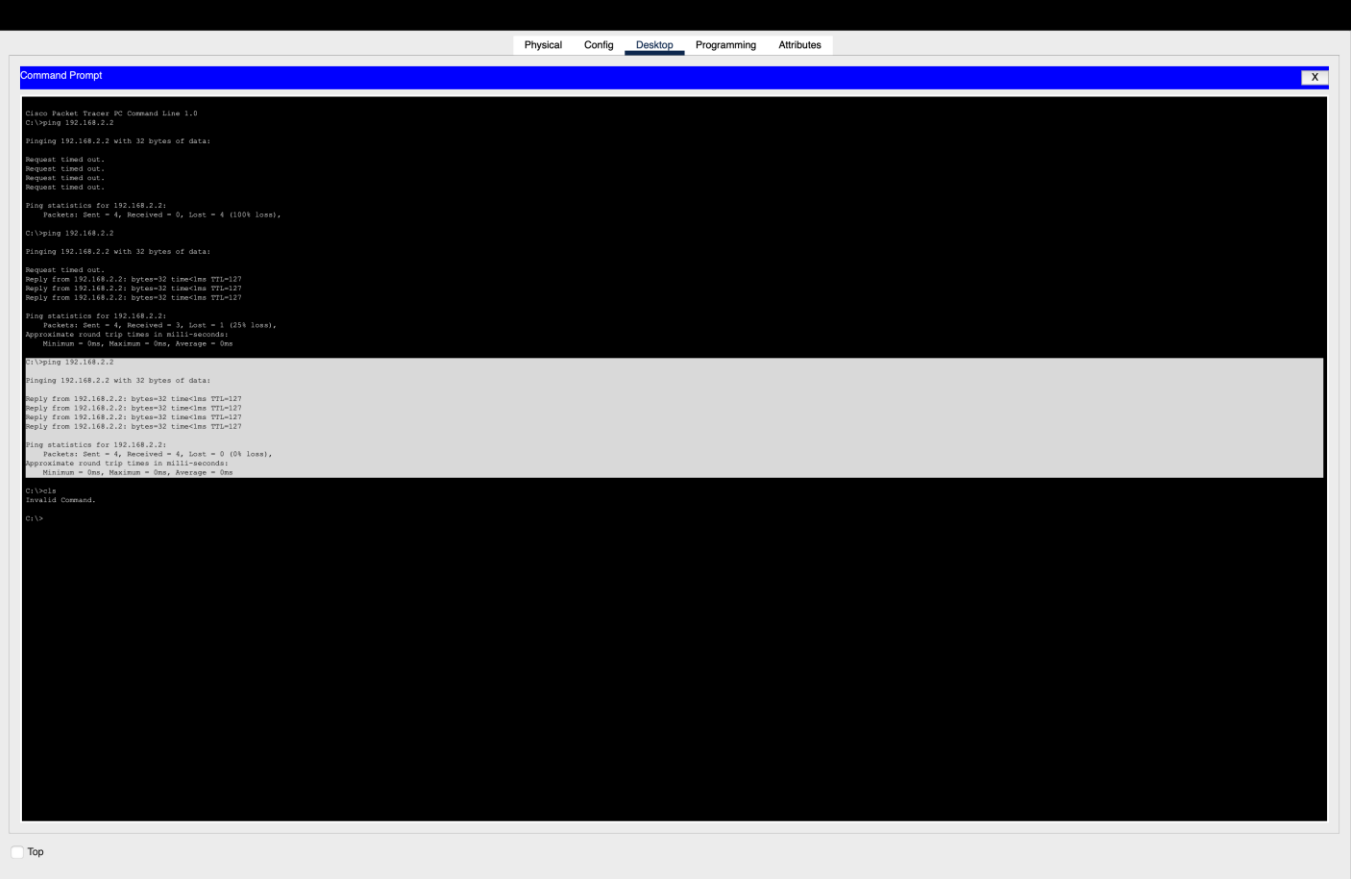


Fig5:RouterInterface Configuration for PC1 Network

Explanation:

The figure shows the configuration of **FastEthernet1/0** on Router0. The interface is enabled and assigned the IP address **192.168.2.1** with subnet mask **255.255.255.0**. This interface acts as the **default gateway for PC1 (192.168.2.2)** and connects PC1's network to the router.

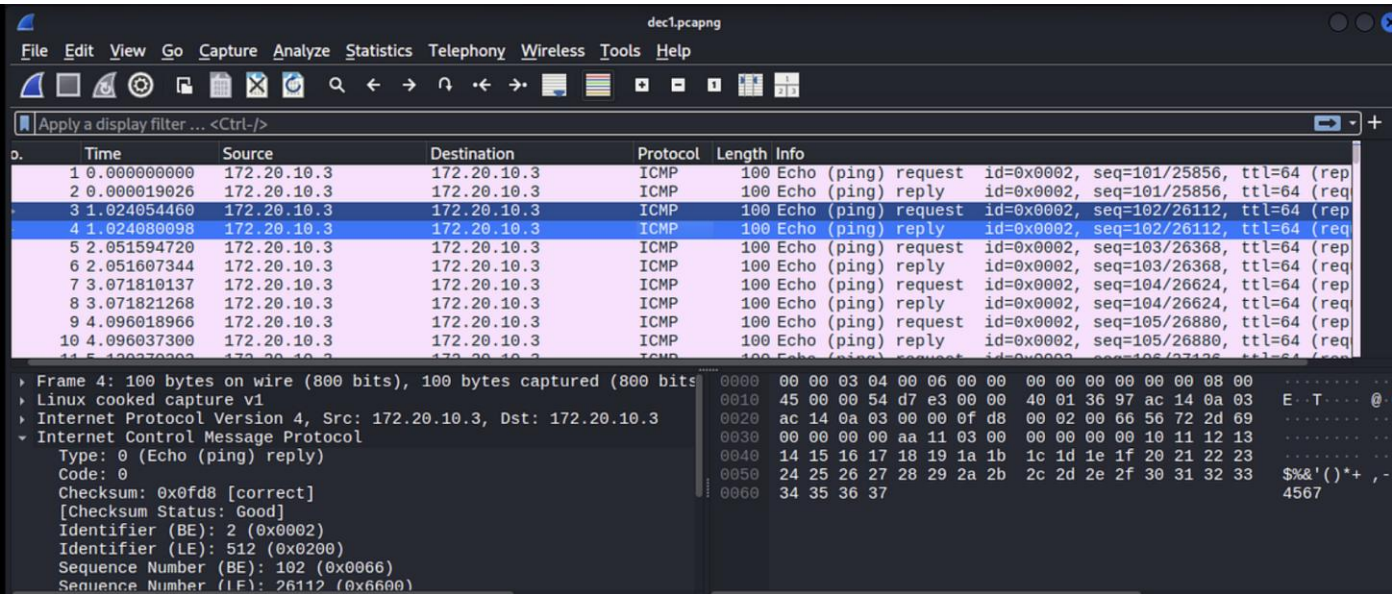


Implementation-6:

Figure 6: Successful Ping Connectivity Between PC0 and PC1

Explanation:

The ping 192.168.2.2 command was successfully executed from PC0 to PC1 through the router. All four ICMP packets received replies, resulting in 0% packet loss. This confirms that the IP configuration and routing between the two PCs are working correctly.



Implementation-7:
Figure 7: Wireshark Capture Showing ICMP Echo Request and Echo Reply Packets

ICMP Field	Echo Request	Echo Reply	Explanation
Type	8	0	Type 8 represents Echo Request and Type 0 represents Echo Reply.
Code	0	0	Code 0 indicates a normal ICMP echo message.
Identifier	Same value	Same value	Identifies the ping process and matches the request with its reply.
Sequence Number	0, 1, 2, 3...	Same number	Identifies the individual ping request and its corresponding reply.

ICMP in Network Troubleshooting and Connectivity Verification:

ICMP helps network administrators check whether devices are reachable and communicating correctly. The ping command uses ICMP Echo Request and Echo Reply messages to verify connectivity between two devices. It also provides information about packet loss and Round Trip Time (RTT), which helps identify network delays or failures. ICMP error messages can indicate problems such as unreachable hosts or networks. Therefore, ICMP is useful for detecting connectivity issues, measuring response time, and troubleshooting network communication.

RUBRICS FOR CALCULATION

Criteria	W	L e v e l 4 – E x e m p l a r y	L e v e l 3 – P r o f i c i e n t	Lev el 2 – Devel oping	Le vel 1 – Be gin nin g
Understandi n g of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technol ogi es	Good understan ding with proper integratio n of most concepts	Basic understan ding ; limited or partial integration	Poor understan ding ; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementatio n with optimal solution	Correct implemen tati on with minor issues	Basic implement ati on with errors or inefficienci es	Incorrect or incomplete implementa ti on
Use of Tools	20	Effective and advanced use of tools/techn ologies with proper configurati on	Appr opriat e use of tools with corre ct execu tion	Limi ted or incon sisten t use of tools	Incorr ect or minim al use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonab le interpret ation	Basic analysis with limited insights	Po or or inc orr ect ana lys is of res ults

Documenta ti on	10	Clear, well- structured documentation with diagrams, code, and results	Organize d documen tatio n with necessar y details	Basic documen tatio n with missing details	Poor or incomple te documen tatio n
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